

Choking Under Pressure: Self-Consciousness and Paradoxical Effects of Incentives on Skillful Performance

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Choking under pressure is defined as performance decrements under circumstances that increase the importance of good or improved performance. A model for choking on coordination and skill tasks is proposed, holding that the pressure increases the conscious attention to the performer's own process of performance and that this increased conscious attention disrupts the automatic or overlearned nature of the execution. Six experiments provided data consistent with this model. Three studies showed that increased attention to one's own process of performance resulted in performance decrements. Three other studies showed similar decrements produced by situational manipulations of pressure (i.e., implicit competition, a cash incentive, and audience-induced pressure). Individuals low in dispositional self-consciousness were shown to be more susceptible to choking under pressure than those high in it.

"Choking under pressure" is a metaphorical expression used to describe the occurrence of inferior performance despite individual striving and situational demands for superior performance. *Pressure* may be defined as any factor or combination of factors that increases the importance of performing well on a particular occasion. *Choking* refers to performance decrements under pressure circumstances. The terms are most commonly used in connection with competition, but their usage does extend to noncompetitive performances.

Although pressure per se has received little direct research attention, research on related topics suggests a model of how it could lead to choking. Competition, which is one type of pressure situation, is generally regarded as arousing. Wankel (1972) identified rivalry, audience presence, and the presence of coactors as the three motivational components of competition, and each of these has been associated with increased arousal and drive (Martens & Landers, 1972; Paulus & Cornelius, 1974; Paulus, Shannon, Wilson, & Boone, 1972;

Steigleder, Weiss, Balling, Wenninger, & Lombardo, 1980; Steigleder, Weiss, Cramer, & Feinberg, 1978; Zajonc, 1965). Wegner and Giuliano (1980) provided evidence that arousal helps focus one's attention on oneself. Indeed, the presence of an audience has been shown to be a cue that focuses attention on oneself (Carver & Scheier, 1978). Finally, Kimble and Perlmutter (1970), Langer (1978), and Langer and Imber (1979) have noted that heightened self-awareness may disrupt the performance of well-learned or "overlearned" skills. More generally, Martens and Landers (1972) suggested that focusing evaluative attention on the *process* of performance may impair the performance more than focusing such attention on the performance *outcome*. Their results were not confined to an overlearned task. Thus, competition is arousing; arousal heightens self-consciousness; and self-consciousness disrupts performance of some tasks.

The present research was based on a model holding that pressure increases self-consciousness, possibly by means of heightened arousal, and that this focus of attention disrupts skillful performances. Under pressure, a person realizes consciously that it is important to execute the behavior correctly. Consciousness attempts to ensure the correctness of this execution by monitoring the process of performance (e.g., the coordination and precision of muscle movements); but consciousness does not contain the knowledge of these skills, so

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that it ironically reduces the reliability and success of the performance when it attempts to control it. To use the familiar example of typing (Langer & Imber, 1979), the attempt to ensure accuracy by conscious monitoring of one's finger movements (necessitated, perhaps, by the pressure of having run out of correction fluid) is often counterproductive.

It must be emphasized, however, that choking is not restricted to well-learned or "over-learned" skills. Past research has defined the distinction between learning and performing in several different ways. One can call it *performing* when the learning has asymptoted, that is, stabilized. One can call it *performing* when the correct response has become dominant (see especially Zajonc, 1965). Finally, one can call it *performing* when the situation defines it as such (instead of as *learning*). The present research is based on the third understanding of the distinction between learning and performing. In other words, the distinction between learning and performing is understood here as determined by the structure of the situation and the intentionality of the subject, not by the intrinsic level of the performance. Although this differs from the traditional (first and second) views, it has some practical justification. A high school athlete or introductory-level student cannot be said to have mastered anything, but such a person will be quite sensitive to the difference between practice and competition (or test), and he or she may certainly choke. My analysis of the concept of pressure emphasizes that it focuses on a particular trial. On that trial, the subject tries to do the best he or she can; learning is incidental. I use *choking* to describe a failure to perform up to whatever level of skill and ability the person has at that time.

An alternative derivation of the present model can be based on Sartre (1974/1934), who argued that human behavior is free and not subject to causal determinism. His reasoning and examples apply only to conscious choices. Consciousness, in that view, frees behavior from certain causal factors; consciousness reduces the automatic or instinctive quality of behavior (cf. also Kimble & Perlmutter, 1970; Langer & Saegert, 1977). Hefferline, Keenan, and Harford (1959) showed that the operation of the laws of operant conditioning on a covert muscle response could be under-

mined by making the person increasingly conscious of the behavioral process by which reinforcement was obtained even when the individual was consciously trying to help (facilitate) the process. Insight psychotherapy presumably relies on a similar principle: By making unconscious material conscious, the therapist frees the neurotic of its determining effects on his or her behavior. Thus, there are several indications that increased consciousness makes behavior less lawful and less predictable. In skillful performance, however, the optimum is for behavior to proceed reliably as a causally ineluctable sequence of events. To the extent that consciousness undermines this causal reliability, consciousness would tend to be detrimental to performance, as in the Hefferline et al. study.

The role of self-consciousness in this model brings up an issue that is central to Experiments 3, 4, and 5, namely, dispositional self-consciousness. If self-consciousness is bad for performance, then in the absence of pressure persons who are habitually low in self-consciousness should outperform those who are high in it. However, if pressure causes self-consciousness, then pressure may affect persons differentially as a function of trait self-consciousness. Persons who are habitually self-conscious should find it easier to cope with situations that engender self-consciousness because they are accustomed to performing while self-conscious. Thus, the increment in self-consciousness caused by going from a non-pressure to a pressure situation may be greater for a person who is dispositionally low (rather than high) in self-consciousness, and it may be reflected in greater choking among persons low in self-consciousness rather than high in it.

The measure of dispositional self-consciousness (Fenigstein, Scheier, & Buss, 1975) contains subscales for private and public self-consciousness. The present investigation was oriented toward using full-scale self-consciousness scores because the theory was formulated in terms of self-consciousness in general and because on an a priori basis it seemed plausible that both subscales could contribute to choking. Specifically, the increased awareness of one's performance process seems to denote a private self-consciousness, but public self-consciousness may increase one's sensi-

tivity to some situational manipulations of pressure (e.g., audience or coactor presence). Results of the present studies therefore emphasize the full-scale self-consciousness scores but will report the effects of the subscales as a way of getting additional insight.

The six experiments here are organized according to the following plan. Experiments 1, 2, and 3 sought to verify that performance decrements are caused by increased awareness of the performance process. Experiments 4 and 5 sought to show that pressure causes performance decrements (i.e., choking)—indeed, decrements similar to the ones caused by the attentional manipulations in Experiments 1, 2, and 3. Experiments 4 and 5 are therefore the most important ones, for they are concerned with the situational and dispositional causes of choking. Finally, Experiment 6 sought to verify that choking is an actual decrement in performance (by using a within-subjects demonstration of choking), to show that it occurs among people who are already good at the task, and to demonstrate choking in a field setting.

The same task was used in Experiments 1 through 5. (Experiment 6 used a popular video game instead.) The task was chosen as an analog to the skillful aspects of athletic performance. Other factors that aid athletic performance (strength, reaction time, stamina) are largely irrelevant to this task. It is described prior to Experiment 1.

Task and General Procedure

The task used in the first five experiments was the commercially available "roll-up" game. A picture and description of the apparatus were given by Martens and Landers (1972). Two rods are attached to a vertical board at one end and the subject holds the other ends in his or her hands. A metal ball rests on the rods. By moving the rods horizontally, the subject attempts to cause the ball to roll along the rods, which is *up* a slight incline. Points are scored by moving the rods apart so that the ball drops into one of the holes in the platform beneath the rods. The object is to roll the ball as far as possible from the starting point (the vertical board to which the rods are attached) toward the subject and to drop it in the hole farthest from the starting

point. The task thus requires motor and visual-motor coordination.

Subjects practiced the task for 5 min. The experimenter came over and observed the final minute, recording the subject's total score for that minute. This constituted the subject's "practice trial" score. The manipulations were then introduced, and the subject performed two 1-min trials. Subjects were always informed of what they had scored.

Experiment 1 (Pilot)

Half of the subjects were told to be aware of their hands while performing the task. The rest heard similar instructions and explanations regarding the ball. The model predicted that increased attention to performance process ("hands" condition) would result in poorer performance than in the "ball" condition. This prediction was supported. Analysis of covariance using baseline (practice trial) scores as the covariate revealed a significant difference, $F(1, 21) = 8.20, p < .01$.

Experiment 1 provided results consistent with the hypothesis that skillful performance is disrupted by increased awareness of one's movements and efforts. If pressure produces similar attentional shifts, then these shifts could help account for choking.

The lack of a no-manipulation control group in Experiment 1 made an alternative explanation plausible—specifically, that performance was improved by the instructions to be aware of the ball. Experiment 2 was designed to rule out this alternative interpretation.

Experiment 2 (Pilot)

Experiment 2 substituted a control group, in which there was no manipulation of awareness, for the ball condition in Experiment 1. The subjects in the hands condition tended to perform worse than the control subjects, but the result did not reach significance, $F(1, 24) 2.00, p = .17$. Still, the mean of the control subjects was quite close to the mean of the ball condition in Experiment 1. Taken together, Experiments 1 and 2 seem to indicate that increased awareness of one's internal performance processes can hamper skillful performance.

Experiment 3

Experiment 3¹ replicated Experiment 2 with the addition of a measure of dispositional self-consciousness (Fenigstein et al., 1975). The implication of Experiments 1 and 2 was that directing attention to performance process could disrupt performance; it seemed plausible that people who are habitually unaware of their internal states and processes (i.e., those low in dispositional self-consciousness) would be especially vulnerable to the deleterious effects of the instructions to attend to these processes.

Method

Subjects. Twenty-five undergraduate students from Case Western Reserve University participated in the experiment. The data from 1 subject were discarded because a brief discussion of performance pressure between her and the experimenter took place just before her performance.

Procedure. The procedure replicated that of Experiment 2, except that subjects filled out a questionnaire measure of self-consciousness (Fenigstein et al., 1975) prior to the practice period.

Results and Discussion

Analysis on the performance scores (combined for both trials), using practice scores as covariate, revealed a significant main effect for focus of attention, $F(1, 19) = 8.51, p < .01$. Other effects were not significant. Simple analysis of variance (ANOVA) on these performance scores similarly showed a significant main effect of attentional focus, $F(1, 20) = 5.01, p < .05$; the main effect for level of self-consciousness and the interaction between focus and self-consciousness were both marginally significant, unlike in the analysis of covariance. Table 1 shows the adjusted means from the covariance analysis.

The disruption of performance by the attentional instructions appeared most clearly in the subjects low in self-consciousness, $t(19) = 2.97, p < .05$, for the adjusted means. Among subjects high in self-consciousness, the effect of the instructions was weak, $t(19) = 1.32, ns$.

The subscales for public and private self-consciousness were scored and subjects were classified as high or low in each of these by median splits. A $2 \times 2 \times 2$ analysis of covariance (ANCOVA) using both subscales as independent variables revealed three significant

Table 1

Mean Performance Scores: Experiment 3

Self-consciousness	Focus	
	Hands	Control
High	49.4	59.6
Low	52.2	71.7

Note. The numbers are adjusted means from analysis of covariance. High scores represent good performance.

effects. The main effect for focus of attention obtained, $F(1, 15) = 10.57, p = .005$. A main effect for *private* self-consciousness reflected the consistently superior performance of persons low in that trait over persons high in it, $F(1, 15) = 8.15, p = .012$. Finally, an interaction between *public* self-consciousness and focus of attention, $F(1, 15) = 10.75, p = .005$, showed the greater vulnerability of persons low in that trait (than of people high in that trait) to performance decrements caused by the attention instructions.

Experiments 4 and 5 were designed to investigate whether situational manipulations of pressure could effect performance decrements similar to those caused by attentional refocusing in the first three experiments.

Experiment 4

Experiment 4 used Wankel's (1972) three components of competition—rivalry, coaction, and audience—to generate pressure. In this study, the competition and the pressure were implicit; Experiment 5 used an explicit manipulation of pressure.

In competition, one source of pressure is the relative score of the contestants. Pressure was previously defined as the importance of performing well on a particular occasion. Given that the object of competition is to win (i.e., to outperform one's opponent), the score is an important determinant of how well one will have to perform in order to achieve this

¹ The six experiments were performed in the sequence in which they are reported here, except that Experiment 3 was actually performed fifth. The author is grateful to James Pennebaker for suggesting that Experiment 2 be replicated with the inclusion of the Self-Consciousness Scale in order to make that study resemble Experiments 4 and 5, as well as for his other suggestions and comments.

success. If one is far ahead, one can afford some errors without losing; pressure is minimal. If one has only a slight lead, the pressure is increased, although an occasional or minor error will keep the contest still undecided (unless perhaps it is nearly over). Pressure would seem to be greatest if one is slightly to moderately behind. In that situation, one retains the possibility of success only if one performs very well; any further mistakes or setbacks may end one's chance of winning. If one is far behind, pressure is presumably diminished. Performing well becomes irrelevant when one has already effectively lost the contest. (This reasoning is consistent with the findings of Seta, 1982).

The present experiment selected two of these situations to compare with a control (no competition) condition. Subjects performed the task in alternation with a confederate who did either moderately better or moderately worse than the subject. It was predicted that pressure would be greater and choking would be more common in the former condition than in the latter.

Rather than have the subject and the confederate compete explicitly for some prize or reward, the present study relied on self-presentational concerns to generate implicit pressure. The contribution of self-presentational concerns to pressure derives from the individual's belief that others will evaluate him or her unfavorably if he or she performs poorly (Baumeister, 1982).

Method

Subjects. Forty-five male undergraduates at Case Western Reserve University participated in the experiment in connection with general course requirements. Three subjects turned around to watch the confederate perform the task; data from these 3 subjects were not included in the analyses. Data from 2 additional subjects were unusable because their unusual form of experimentation during the practice period rendered the baseline (practice trial) data meaningless. Subjects were randomly assigned among treatments, as always.

Procedure. The subject signed up for an experiment on "Personality and Game Performance." Upon arrival, he was given two questionnaires to fill out, which contained the Self-Consciousness Scale (Fenigstein et al., 1975) and the Bem Sex Role Inventory (Bem, 1974).² Subjects were run individually.

Control condition. When control subjects completed their questionnaires, they were moved to the other table and put through the practice period and performance trials just as were the control subjects in Experiment 2.

Pressure conditions. Shortly after the subjects in the experimental condition began working on the questionnaire, there was a knock on the door. The experimenter answered it: A female student (the confederate) stood there. She explained that she had been scheduled to do the experiment that morning but had been unable to show up. After a brief discussion, the experimenter somewhat reluctantly "decided" that it would be feasible to run 2 subjects at once.

The experimenter pretended to decide that the best plan was for the 2 subjects to alternate turns at the roll-up task, each returning to the questionnaire while the other was doing the task. The confederate went first. During the subject's practice period, the experimenter asked him and then the confederate if they had played the game before; the confederate gave the same answer the subject did.

After the subject's practice period, the experimenter gave the instructions for the two performance trials (how the scoring worked, the importance of doing one's best, etc.). Ostensibly to demonstrate scoring, the subject was told what his score "would have been" had the last minute of his practice period been a performance trial. The subject was then returned to his questionnaire (seated with his back to the table with the roll-up) and the confederate performed her first trial.

The experimenter pretended to record her performance. However, when she completed the 1-min trial, the experimenter announced a score for her that was computed based on the subject's practice trial score. Later, the confederate's apparent score on the second trial was computed based on the subject's first trial. In the high-pressure conditions, her scores were 13 and 11 points above what the subject had just scored. In the low-pressure conditions, she apparently scored 7 and 9 points below the subject's previous score.³ Thus, the subject clearly heard that the confederate had scored definitely better or worse than he had. As the subject prepared to start his second trial, the confederate interrupted to say she had finished the questionnaire and to ask permission to watch the subject perform. The experimenter said "Sure," as if it were unimaginable that her surveillance would affect the subject's performance. She sat on the table next to the subject (on the other side from the experimenter). She made no comment or reaction; when the subject finished, she left. The subject then finished his questionnaire and was debriefed.

Results

Manipulation check. During the debriefing, the experimenter asked subjects whether they had noticed the discrepancy between their own performance level and the confederate's.

² The BSRI was included as a possible means of identifying "machos" (ultra-masculine) males who might be unusually susceptible to competition with a female. However, the sample of men in that category was too small to permit a reasonable test.

³ Scores below 5 are simply implausible, so if the subject scored extremely low, the confederate's score was given as closer to it than the formula specified, although of course it was still below the subject's score.

Table 2
Mean Performance Scores: Experiment 4

Self-consciousness	High pressure	Low pressure	Control
High	74.0 (77.5)	80.1 (84.8)	67.2 (70.7)
Low	65.1 (61.3)	83.3 (81.7)	81.2 (74.7)

Note. Numbers in parentheses are for analysis using private self-consciousness instead of the full scale. The other numbers are adjusted means from analysis of covariance. High scores represent good performance.

Nearly all subjects reported that they had. A few said they had paid no attention, but such statements could have been face-saving gestures rather than accurate reports.

Performance data. Table 2 shows the mean (adjusted for covariance) performance scores. Scores are based on the combined scores of the two performance trials. Both measures showed significant interactions between pressure and dispositional self-consciousness. An ANCOVA (using the practice trial scores as covariate) revealed a significant interaction, $F(2, 33) = 3.99, p < .05$, as well as a significant main effect for pressure, $F(2, 33) = 4.47, p < .02$.

The pattern of means largely supported the predictions. Persons low in self-consciousness performed significantly better than those high in self-consciousness in the control condition, $t(33) = 2.40, p < .05$. However, persons low in self-consciousness were the ones who choked under pressure. Persons low in self-consciousness performed much worse in the high-pressure situation than in the control situation, $t(33) = 2.56, p < .02$, whereas persons high in self-consciousness showed a nonsignificant improvement. Under high pressure, persons low in self-consciousness were marginally worse than those high in it, $t(33) = 1.62, p < .12$.

Both high- and low-self-consciousness subjects performed well under low pressure. In fact, highly self-conscious persons performed significantly better under low pressure than they did in the control condition, $t(33) = 2.07, p < .05$. This result suggests that a little situational pressure combined with favorable feedback may facilitate performance. Perhaps when one is successful one does not feel the

need to pay increased attention to one's performance process, so that the self-defeating attentional shift is avoided. Such an explanation is consistent with the finding that persons with low dispositional self-consciousness choked only when they needed to improve in order to avoid being outperformed by the confederate; when they were already doing better than she, these subjects performed quite well, $t(33) = 3.29, p < .005$, for the subjects low in self-consciousness in the two pressure conditions.

The predictions regarding the subscales of the self-consciousness scale were also supported. Private self-consciousness did interact with pressure, $F(2, 33) = 3.68, p < .05$, on the ANCOVA on the combined scores, with practice score as covariate. The pattern of means resembled that of the analysis that used full scale self-consciousness scores (see Table 2). The difference between the performance of the two self-consciousness groups in the control condition did not reach significance when private self-consciousness was used to differentiate the groups. However, the difference between the performance of the subjects who were high and low in private self-consciousness in the high-pressure condition was highly significant, $t(33) = 3.18, p < .01$; subjects high in self-consciousness performed better. Private self-consciousness does seem to be closely related to choking.

The interaction between public self-consciousness and pressure was not significant, $F(2, 33) = 1.41, ns$, as predicted. Public self-consciousness did, however, contribute to predicting the effect of being watched by the confederate while performing the second trial. A repeated-measures ANCOVA revealed a significant interaction between trial, public self-consciousness, and pressure, $F(2, 34) = 9.02, p < .001$. In the high-pressure condition, subjects performed about the same on the first trial regardless of their level of public self-consciousness; but when watched by the confederate on the second trial, persons with high public self-consciousness improved, whereas those with low public self-consciousness got worse. This is consistent with the general interpretation here that situational increases in self-consciousness disrupt performance less among individuals who are habitually self-conscious than among those who are not.

The possibility that the results were mediated by differential effort was explored by considering the number of times the subject let the ball drop as a rough measure of effort. Most subjects typically dropped the ball two or three more times during a performance trial than during the practice, which is consistent with increased effort. However, an ANOVA on the number of times the ball dropped revealed no significant effects, interactions, or differences.

One last finding may deserve mention. As a measure of consistency of performance, the absolute value of the difference between the two performance trial scores was computed for each subject. An ANOVA on these scores revealed a marginally significant main effect for self-consciousness, $F(1, 34) = 2.87, p = .10$, and a significant interaction between (full-scale) self-consciousness and pressure, $F(2, 34) = 8.43, p < .001$. The high-self-consciousness group in the low-pressure condition was exceptional. Their performances were significantly more inconsistent than those of low-self-consciousness persons in the same condition, $t(34) = 3.11, p < .01$. Although this effect was not predicted, it may warrant further investigation.

Discussion

The results of Experiment 4 were consistent with the predictions based on the model of choking as mediated by attention to self (i.e., to one's internal processes involved in the performance). In the absence of situational pressure, high dispositional self-consciousness hampered skillful performance. Persons who were habitually unaccustomed to being self-conscious were, however, the ones who choked (i.e., who showed performance decrements in the situation that called strongly for performance increments). Persons with high dispositional self-consciousness seemed better able to avoid the performance-inhibiting effects of situational pressure.

Several qualifications must be made. The pressure in this situation was implicit; that is, it is only by inference that one can assume greater pressure to perform well when in danger of being outperformed than when performing alone or performing at a level com-

fortably superior to that of the coactor. Experiment 5 was designed to remedy this weakness by using an explicit manipulation of situational pressure.

Moreover, the pressure conditions differed from the control condition in the presence of the coactor. The additional observer may have contributed to social facilitation effects (Zajonc, 1965). Experiment 5 was also designed to avoid confounding pressure with the presence of additional persons in the room.

The negligible correlation ($r = -.08$) between practice trial score and change score (performance minus practice) would seem to disconfirm a suggestion that the results of this experiment may have been influenced by ceiling or floor ability effects. In particular, the Hull-Spence notion that increased drive would facilitate performance for those with high initial skills but inhibit performance for those with poor initial skills was not in evidence.

Experiment 5

The goal of Experiment 5 was to demonstrate choking with an explicit and unfounded manipulation of pressure, in the sense of the importance of performing well on a given occasion. Subjects were offered an incentive in the form of cash if they would perform at a certain level. The level was computed based on the subject's baseline (practice trial) performance according to reasoning similar to that in the high-pressure condition of Experiment 4. That is, it was set within the subject's presumed reach, but he or she would have to perform quite well (for his or her ability level) in order to achieve it. The improvement necessary was the highest mean change score among the six cell means of Experiment 4.⁴

It was predicted that subjects low in self-consciousness would perform worse in the incentive condition than in the control condition, in which no money was used. Based on the results of Experiment 3, a statistically weak trend toward choking was predicted for subjects high in dispositional self-consciousness.

⁴ The highest mean change score in Experiment 4 was 27, which is for two trials. For one trial, then, an improvement of 14 was required to meet the criterion.

Method

Subjects. Thirty-seven male and female undergraduates at Case Western Reserve University took part in this experiment under the same terms as in Experiment 4. (No subject participated in both experiments.)

Procedure. Each subject was greeted by the experimenter and given the same initial explanation as in Experiment 4. The subject was given only the Self-Consciousness Scale to fill out. When he or she completed it, the experimenter asked him or her to move to the performance table. The practice trial was conducted the same as in the previous experiments.

The procedure for the control condition was identical to that of Experiments 2, 3, and 4.

The procedure for the pressure condition diverged from that of the control condition at the end of the practice period. Just before the beginning of the first performance trial, the experimenter took two dollar bills from the pocket of his sport coat and laid them on the table. He said that in order to motivate the subject to perform well, he was offering subjects one dollar for each trial in which the subject's performance exceeded the criterion "that we're using." He named a criterion 14 points above the subject's practice trial score. However, the experimenter acted as if the same absolute criterion was used for all subjects; the subject was not made aware that the criterion had been computed in relation to his own practice score.

The two trials were then performed, with an interval of 1 min between them. Subjects were then debriefed. Subjects who had won money were paid accordingly.

Results

An ANCOVA on the combined performance trial scores, using practice score as a covariate, revealed marginally significant main effects for both variables, $F(1, 32) = 2.62, p < .12$, for pressure, and $F(1, 32) = 3.15, p < .10$, for self-consciousness. The interaction between them was not significant, $F(1, 32) < 1, ns$. The main effects indicated superior performance in the control (no money) condition and for subjects low in self-consciousness, respectively.

Several subjects in the cash (pressure) condition appeared to choke on the first trial but approached or reached the criterion for success on the second trial. When asked about the discrepancy, they said that they had "given up" on reaching the criterion after the failure on the first trial; that is, they had dissociated themselves from the performance pressure by choosing to disregard the incentive. The pressure manipulation may have thus lost some of its impact on the second trial. To explore that possibility, a repeated-measures ANCOVA was conducted, separating the two trial scores.

Table 3

Mean Adjusted Scores: Experiment 5

Trial	Cash		Control	
	HSC	LSC	HSC	LSC
1	27.3	29.4	31.8	35.4
2	32.5	33.9	30.8	37.4

Note. HSC denotes high self-consciousness, LSC, low self-consciousness. The numbers are adjusted means from analysis of covariance. High scores represent good performance.

The main effect for trial was highly significant, $F(1, 33) = 8.05, p < .01$, showing substantially superior performance on the second trial over that on the first trial. That this effect obtained mainly in the cash conditions was confirmed by a significant interaction between trial and pressure, $F(1, 33) = 5.31, p < .05$. Table 3 shows the means.

On the first performance trial, the low-self-consciousness subjects appeared to choke; they performed worse in the cash than in the control condition, $t(33) = 2.26, p < .05$. The high-self-consciousness subjects showed a weaker trend toward choking, $t(33) = 1.51, .10 < p < .15$. The predictions were thus supported for the first trial. They did not receive support on the second trial.

Discussion

Subjects of both levels (high and low) of self-consciousness showed signs of choking in Experiment 5, although the effect was stronger for those low in self-consciousness. Thus, offering a reward for improved performance paradoxically caused subjects to perform worse.

The fact that choking appeared on the first trial and not the second suggests some caution in interpreting the results. However, if the subjects' self-reports were accurate, they lend further support to the model. If "giving up" on reaching the criterion resulted in improved performance, then the pressure of the criterion was indeed a factor that was hampering performance. The fact that subjects could avoid the effects of pressure by internally abandoning the goal also implies that the situation alone does not create pressure—rather, some internal

acceptance of the importance by the performing individual may be necessary in order for choking to occur. The possibility that a private cognitive reorientation by the individual may help prevent choking calls for further research.

Experiment 6

Experiment 6 was an attempt to provide a clear demonstration of choking in a field setting on a reasonably well-learned task. Although Experiments 1 through 5 showed that various situational and dispositional factors affect performance levels, two facts suggest reservations about interpreting them. First, the task was new to some subjects, and 5-min practice may not have overcome the novelty. Second, there was no proof that each individual subject who supposedly choked could actually have performed better in a different situation. Practice trial scores were sometimes higher than performance scores, but the cell means rarely reflected an average decrease in score from practice level. Although the weight of the between-subjects analyses suggests that subjects who choked would have done better in different circumstances, it is still somewhat plausible that the manipulations facilitated performance in some conditions instead of hampering it in others. Experiment 6 was designed to alleviate these interpretive problems.

Method

Subjects. Subjects were customers at a video game arcade in a shopping mall. They were recruited by being offered a free game at the game they were playing. Potential subjects were excluded on the basis of any of the following: failure to reach a minimum cutoff score in their own ("practice") previous game, which signified a lack of ability at the task; failure to have played the same game on at least several previous days (in answer to direct questioning); or age below 13. Thirteen subjects took part.

Procedure. Potential subjects were observed surreptitiously while playing either of two popular video games, "Pac Man" or "Ms. Pac Man." If the subject reached the criterion score, the experimenter approached the subject upon completion of his or her game and said, "I'd like to give you a free game, if you have a minute. I'm doing research on these video games." No subject declined to participate. The experimenter verified that the subject had played the game on previous occasions.

Pressure was manipulated by self-presentational concerns. The experimenter asked the subject's name and age and recorded these. (The experimenter also quietly recorded the subject's previous score, which was displayed on the screen, and which served as a "practice" score.) The experimenter then said "I want you to get the best score you

possibly can, OK? I can only give you one chance, so make it the best you can do." The experimenter then observed the subject's performance, pretending to time it with a stopwatch. The subject's score was recorded and the subject was thanked.

Results

Percentage of change scores were used in this study because of the interest in demonstrating actual decrements in performance and because the two different games gave different levels of scores. Therefore, for each subject, the adjusted change score equaled the performance score minus the practice score, divided by the practice score.

The average change was a 25% drop in performance from the previous (practice) trial to the pressure (performance) trial. This was significantly below zero, $t(12) = 2.91$, $p < .02$, indicating that it was a definite decrement in performance.

Discussion

A verbal instruction emphasizing the importance of doing well on that particular trial, combined with enhanced self-presentational concerns created by getting the subject's name and by overt surveillance of the subject's performance, resulted in performance inferior to what the subject had just done on the same game. In Experiment 6, situational pressure clearly resulted in choking.

The use of a performance criterion ensured that all subjects were good at the task. Indeed, the practice scores of the subjects were from three to seven times as high as those of obvious beginners. Although the subjects perhaps cannot be said to have totally mastered the game (in part because total mastery of one of these games is hard to define: No one stays on forever), they all can be said to have achieved substantial skill prior to the experiment.

General Discussion

The present series of experiments was based on a model of choking as mediated by attentional shifts. According to this model, situational demands for excellent performance (i.e., pressure) cause the individual to attend consciously to his or her internal process of performance, and this consciousness disrupts that process and harms the performance. The re-

sults of the experiments were generally consistent with that model. Performance of a skill task was lowered by directing the individual's attention to his or her internal performance process and by dispositional self-consciousness. Similar performance decrements were obtained by competitive pressure and by a cash incentive offered for improved performance.

Experiments 1, 2, and 3 suggested that self-consciousness harms performance. Experiments 4, 5, and 6 suggested that pressure harms performance similarly. The present experiments did not provide direct evidence that pressure causes increased attention to one's own internal performance process, although there was ample indirect evidence of relationships between pressure and such self-consciousness. Further research is needed to substantiate that relationship.

This model would not apply directly to performance based on strength, stamina, or perseverance because such performances are not based on the execution of automatically learned response sequences. However, self-consciousness could mediate choking on such tasks by increasing the person's awareness of bodily fatigue, pain, and stress, so that he or she decides to reduce the effort. Pennebaker and Lightner (1980) provided results suggestive of that hypothesis, although their subjects were never instructed to try to perform well.

The relevance of self-consciousness to impaired cognitive performance has also been suggested. Wine (1971) cited the focus of attention on oneself as an important feature of test anxiety. In persons high in test anxiety, reduction of "self-preoccupation" has been associated with improved performance (e.g., Sarason, 1981). It is plausible that the performance decrements of test-anxious individuals follow from processes similar to those that mediated choking in the present experiments.

Moreover, the relation of performance expectancies to choking under pressure deserves investigation. Past research has suggested that expectancies can help bring about poor performance (Aronson & Carlsmith, 1962; Baumeister, Cooper, & Skib, 1979; House & Perney, 1974; Zanna, Sheras, Cooper, & Shaw, 1975). Zanna et al. suggested that the combination of internal and external expectancies of success could backfire and lead to choking. It is plausible that the performance decrements

in Experiments 4 and 5 (although not those in Experiments 1, 2, and 3) were mediated by expectancies formed by the subject during the experiment and never expressed overtly. Such an explanation would, of course, need to account for the influences of dispositional self-consciousness on performance, either by differential expectancy formation or differential impact of expectancies as a function of self-consciousness.

A final possible interpretation deserves mention. It could be suggested that attention to oneself decreases one's attention to the task, so that performance decrements are due to overlooking information necessary to perform the task. Essential to this interpretation is the view that self-consciousness and attention to anything else (other than oneself) are mutually exclusive. This view was suggested by Duval and Wicklund (1972) but contradicts other theorists who have regarded self-awareness as superimposed on awareness of external circumstances (e.g., Kant, 1956/1787). This controversy awaits definitive empirical resolution. Still, it must be noted that performance decrements cannot be equated with lack of awareness of task matters; Hefferline et al. (1959), Kimble and Perlmutter (1970), and Langer and Imber (1979) have all indicated that increased attention to certain tasks can impair performance.

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